• Easy • #67 • Add Binary

▲ **Problem Statement:** Given two binary strings "a" & "b", return their sum as a binary string.

▲ **Main Problem Statement:** Add the two binary strings from right to left, just like normal addition.

Keep track of a "carry" :-

- Sum = digit A + digit B + carry
- Current binary digit = Sum % 2
- New carry = Sum / 2

Since the strings can can have different lengths, continue until both strings & the carry are finished.

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Brute Force Method → Convert both binary strings into decimal integers, add them, & convert the result back to binary.

This can cause integer overflow because the strings can contain up to (10^4) characters.

Algorithm → 1| • Start pointers "i" & "j" at the ends of "a" & "b"

2| → Initialize "carry = 0"

3| • Take the current digits from both strings.

4| • Add them with "carry"

5| • Store "sum % 2" in the result

6| → Update "carry = sum / 2"

7| → Move both pointers backward

8| → Reverse the result because digits were added from right to left.

9| • Return the result.

Important Points → • Start from the rightmost digits

• '0' & '1' are characters, so use '-'0'' to convert them to integers.

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- "Sum % 2" gives the current binary digit
- "Sum / 2" gives the carry
- Reverse the result at the end.
- No need to convert the entire binary string to an integer.

▲ Complexity - Time → $O(\max(n, m))$
Space → $O(\max(n, m))$ for the result.